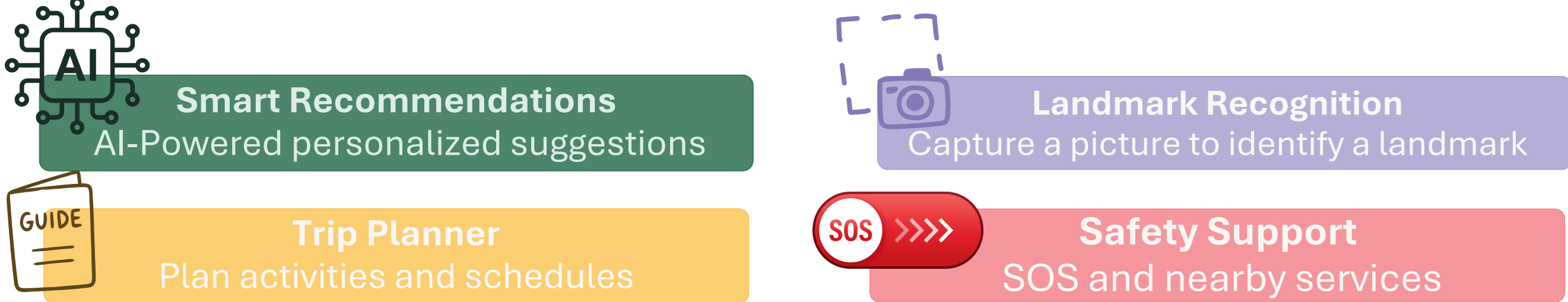


ABSTRACT

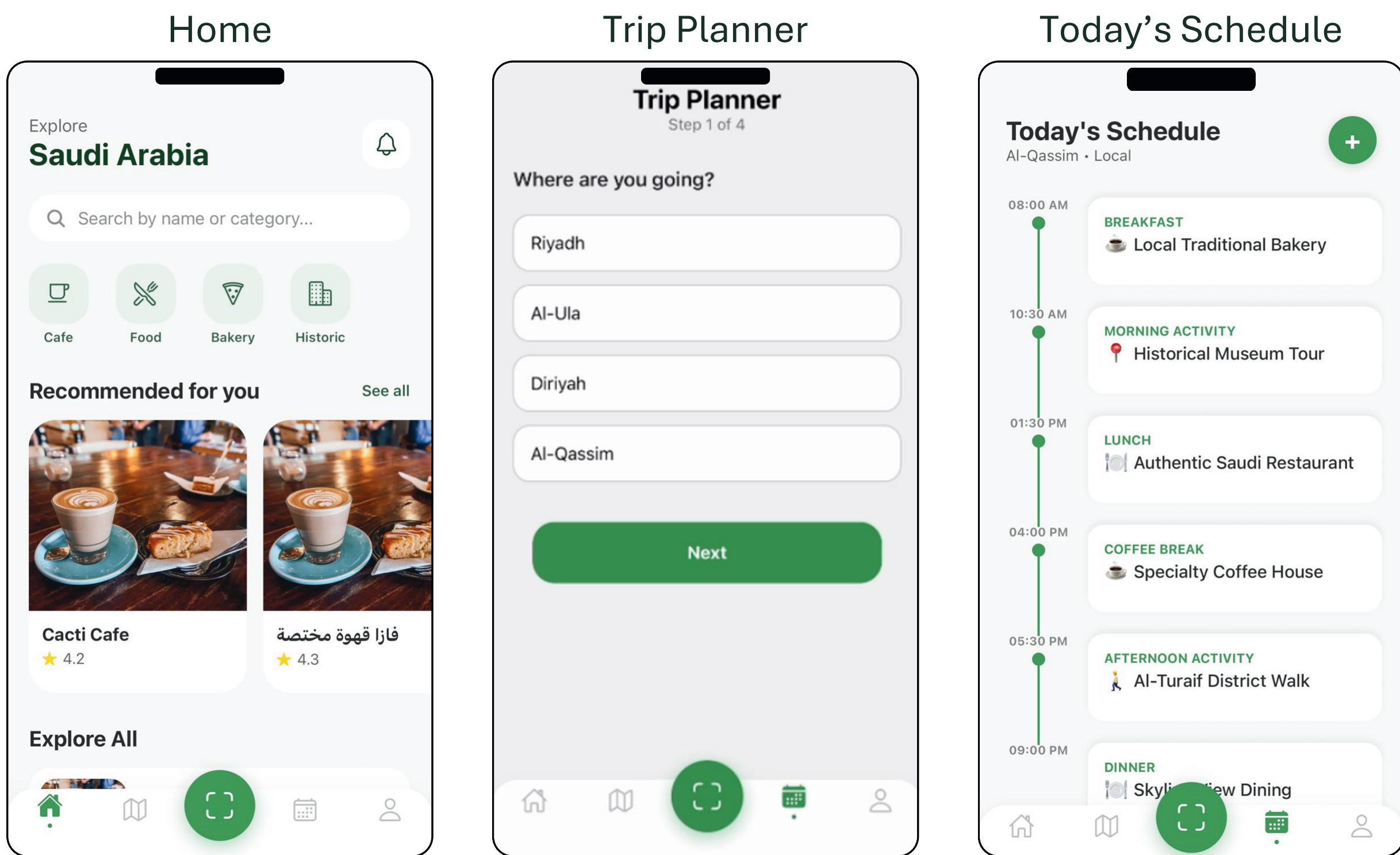
Terhal is an AI-powered tourism mobile application designed to support international tourists in Saudi Arabia. The app helps users overcome language barriers, discover personalized destinations, recognize landmarks, plan trips, and access safety services. By integrating recommendation systems, NLP, image recognition, maps, and emergency features, Terhal provides a safer, smarter, and more personalized travel experience within the Kingdom.

INTRODUCTION

Saudi Arabia’s tourism sector is rapidly growing as part of Vision 2030, attracting increasing numbers of international visitors and expanding cultural, leisure, and event-based tourism. However, many tourists still face challenges such as language barriers, difficulty accessing cultural information, limited personalized guidance, and the need for quick safety support. To address these challenges, Terhal is proposed as an AI-powered mobile application that combines multilingual support, landmark recognition, and the provision of historical information about landmarks, personalized recommendations, trip planning, and safety features to create a smarter and more convenient tourism experience in Saudi Arabia.



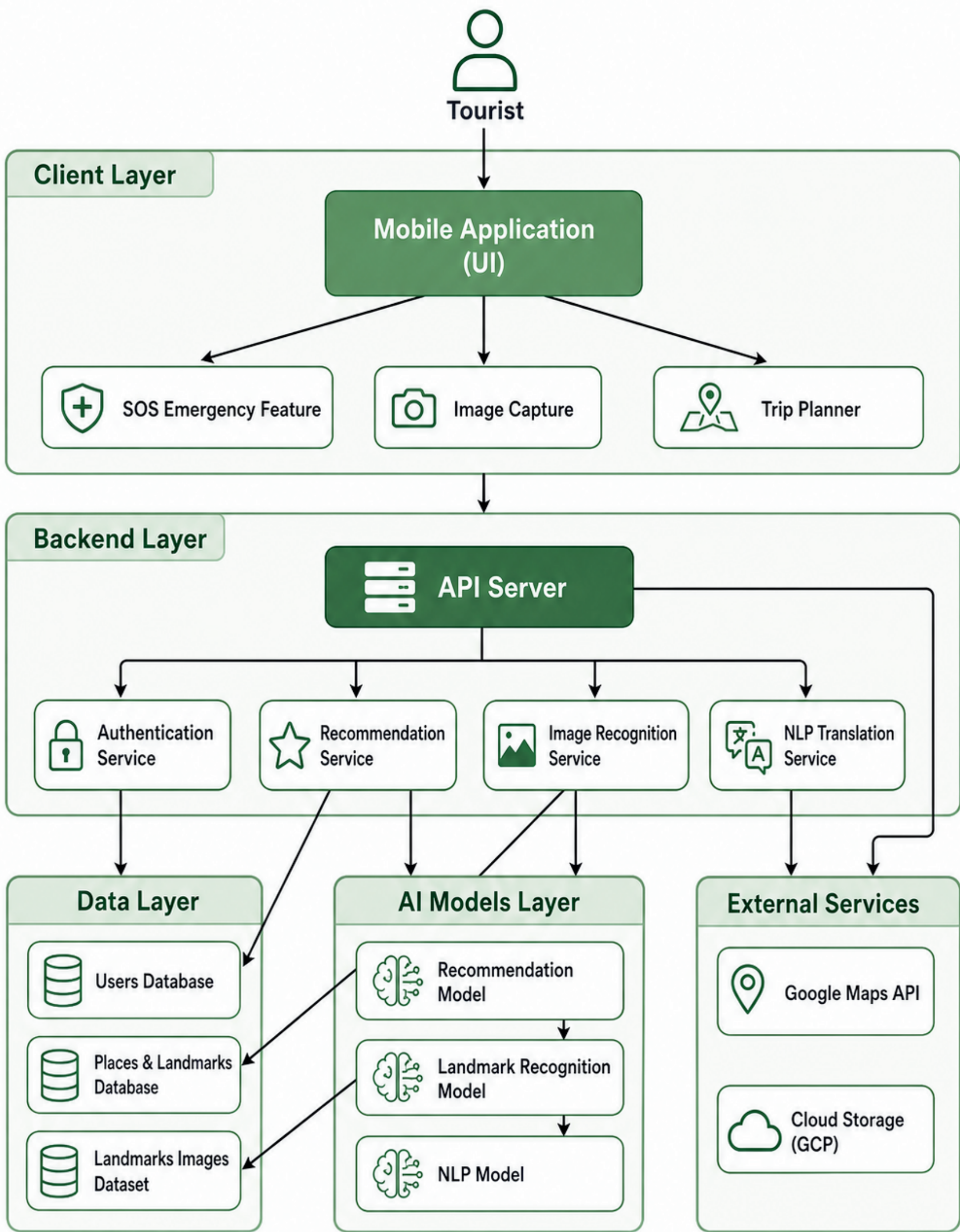
INTERFACE



METHODOLOGY

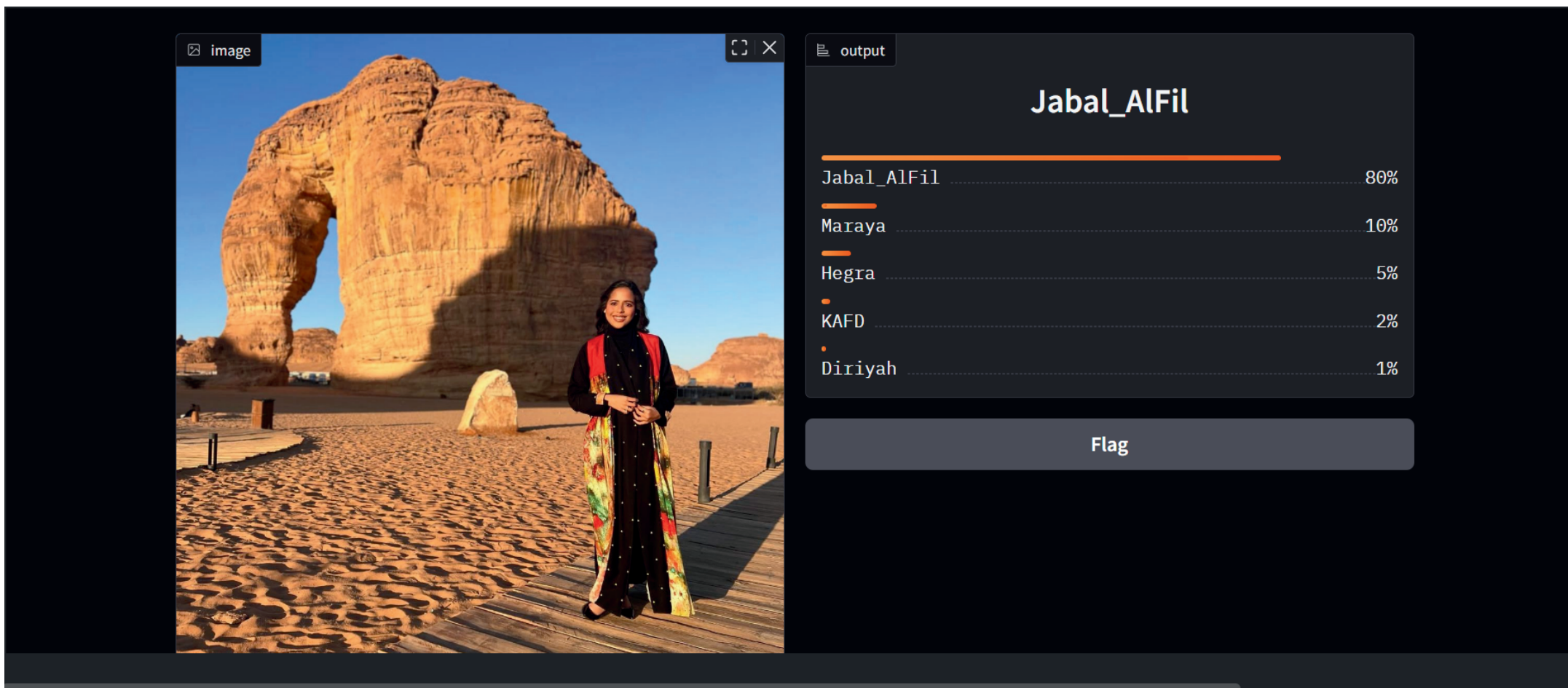
The development of Terhal followed a structured methodology that included data collection and preprocessing, interface design, backend development, and the integration of AI-based services. The system architecture consists of a mobile application connected to an API server, which manages core services such as authentication, recommendation, image recognition, and NLP translation. These services interact with databases, AI models, and external tools such as Google Maps and cloud storage to provide a seamless tourism experience.

Terhal Application Architecture Diagram



AI MODELS RESULTS

MODEL	PURPOSE	METHOD	METRICS & RESULTS
Recommendation System	Places recommender	Hybird Item-Based Collaborative Filtering	HR@10=0.8242 NDCG@10=0.7787
Landmark Recognition	Saudi landmark classification	EfficientNet-B0 Deep Learning	Accuracy=89.81% F1-score=0.90



Simple Gradio interface to test Landmark Recognition Model

CONCLUSION

Terhal provides an AI-powered tourism solution that supports personalized exploration, cultural engagement, and tourist safety. The strong performance of the recommender and landmark recognition models shows Terhal’s potential to enhance international tourism in Saudi Arabia.



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